



Endogenous stimulation-driven Janus mesh with antibacterial, anti-adhesive and prohealing performances for hernia repair

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ABSTRACT

Hernia incidence remains a high level worldwide as witnessed by the fact that more than 20 million hernias are estimated to be repaired each year. Mesh-based hernioplasty allowing wound defects to close more easily and low recurrence rate is considered as the gold standard option for hernia repair. However, clinical polypropylene mesh (PPM) is still lack of effective treatments for the unavoidable bacterial infection, postoperative adhesion and foreign body reaction. Herein, a stimulus-responsive gel was constructed on the PPM by a simple photo-initiated polymerization. The homogeneous gel-modified mesh could spontaneously transform to a Janus mesh locally under endogenous stimulations evoked from inflammatory response or bacterial infection, and demonstrate asymmetric performances. One side of the Janus mesh facing the abdominal viscera could maintain intact gel structures and form a hydration layer to hinder the stick of proteins and cells, while the other side adjacent to the hernia defect site where bacteria prefer to colonize would accelerate degradation due to bacterial growth and release the bactericidal agent for combating the bacterial infection. The in vivo experiments' results demonstrated that no adhesions were detectable on the Janus mesh treated group on the 14th day and the level of inflammatory response in histological evaluation was lower as compared to the commercial PPM. Overall, the endogenous stimulation-driven Janus gel network in abdominal wall wound provides a new approach for soft tissue repair biomaterials.

1. Introduction

Nowadays, hernia has become a frequently encountered issue worldwide as witnessed by more than 20 million people requiring hernia surgery each year [1–3]. Polypropylene mesh (PPM) taking advantages of non-cyto/genotoxicity, structural stability and biomechanics-matched property has become the most widely used material for hernia repair clinically [4,5]. In comparison with the traditional suture-based herniorrhaphy, the PPM based hernioplasty rendering wound defects with easier closure, tension-free, and mechanical strength is recognized as the golden standard option [6–8]. Nevertheless, this “gold standard” still causes not-so-trivial postoperative complications, especially serious abdominal adhesion induced singly or in combination by the surgical trauma, foreign body reaction, persistent inflammation [9–12]. Additionally, hernia mesh-related bacterial infection is a

frequent side effect of abdominal surgery which could amplify the inflammation response and aggravate the abdominal adhesion [13,14].

Currently, intensive strategies have been proposed to modify meshes for limiting the occurrence of implant surgery related bacterial infection and visceral adhesion. One of the most obvious solutions is to cover the whole surface of the mesh using a highly hydrophilic coating to create a physical barrier that can directly separate the implanted material from adjacent normal organs and suppress the initial bacterial adhesion on the mesh surface [15–18]. Indeed, after the introduction of hydrophilic coating on the surface of PPM, proteins responsible for cellular adhesion and bacterial cells do not have affinity to the surface as well as fail to trigger the adhesion formation between the abdominal viscera and the PPM [13]. However, this whole surface or double-sided antiadhesion modified strategy can also prevent the adhesion and proliferation of soft tissues, which is undesirable for the tissue regeneration and PPM

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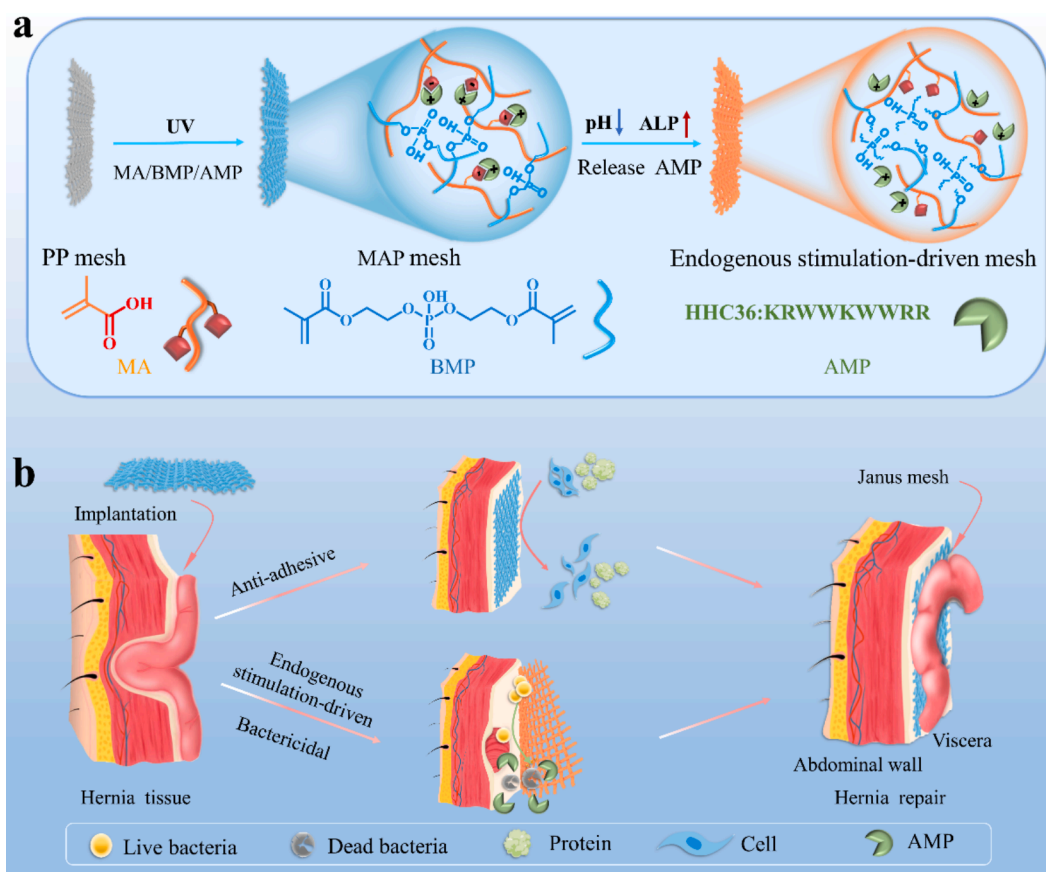
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integration [6]. In this regard, antiadhesive strategies focus only on physical barriers might prolong the foreign body response and the inflammatory reaction, and thus severely compromise their repair consequences [14,19,20]. Hence, achieving host cells adhesion and proliferation on the mesh to form a rapid and healthy host integration while avoiding the adhesion with adjacent tissues is the key for the successful hernia repair. Nevertheless, the realization of anti-adhesion of viscera and promotion for integration is often based on contradictory tissue anti-adhesion and adhesion behaviors, which is difficult to simultaneously achieve based only on a single mesh.

Recently, inspired by the asymmetric performances of peritoneum, the construction of Janus hydrogel which could harmoniously integrate the completely opposed properties (such as anti-adhesion and pro-healing) into one single material has gained great attentions in the field of internal soft-tissue defect repair [21–24]. For instance, Liang et al. developed a novel asymmetric porous Janus hydrogel whose densely porous and smooth bottom-surface could effectively suppress the visceral adhesion, meanwhile the loose extracellular-matrix-like porous and rough top-surface can significantly improve fibroblast adhesion and tissue growth [24]. Cui et al. reported an interesting phenomenon that based both on the component of polyelectrolyte hydrogel containing ionic or non-ionic carboxyl groups could demonstrate strikingly distinct adhesive and non-adhesive [25]. Although efficacy has been shown, the practical application of these as-prepared Janus hydrogel meshes is still fairly limited due to that: (i) the operational process is commonly cumbersome and frequently requires harsh organic solvents which is unsuitable for large-scale preparation and might cause biocompatibility problems [24]; (ii) the mechanical property of hydrogel meshes could hardly match with intra-abdominal

pressure, and the mismatched transmission of loads at the meshes-tissue interface would lead to hernia recurrence [6]; (iii) repair with surgical meshes has been considered as the first choice when the wall defect was more than 2 cm [26]. In this context, the introduction of the asymmetric biological properties into the most commonly used PPM to construct a “Janus PPM” equipped with antiadhesive, host integrated and appropriate biomechanics-matched properties might be an ideal strategy for hernia repair [27,28]. Several attempts were made to fulfil the different requirements on each side in hernia mesh. For example, Tanushree et al. reported a side-specific PP mesh with one side grafted with poly (2-methacryloyloxyethyl phosphorylcholine) (PMPC) for anti-adhesion to protein and the other side coated with chitosan and functionalized nanodiamond (Chi/FND) for promoting cell attachment and proliferation [27]. Mahvash et al. modified each side of PP mesh using electrospun mats with anti-adhesion and antibacterial properties on the front and adhesion properties (reduce undesired displacements) on the backside [28]. However, the preparation of Janus PPM is complex and the biological properties of the meshes require further investigation in vivo.

Herein, we propose and develop a novel methodology to equip the polypropylene mesh with Janus biological property. As illustrated in Scheme 1, a multi-responsive gel coating composed of pH sensitive methacrylic acid (MA), phosphatase sensitive bis[2-(methacryloyloxy) ethyl] phosphate (BMP), and antimicrobial peptide (AMP) was firstly constructed coaxially on the fiber of PPM to form a homogeneous antifouling and bacterial killing armour through one-step photo-initiated graft polymerization method. The as-constructed homogeneous PPM could spontaneously develop to a Janus polypropylene mesh (JPPM) stimulated by the asymmetrical microenvironments adjacent to



Scheme 1. (a) Schematic illustration of fabrication for an endogenous -responsive Janus gel mesh by one-step photo-initiated graft polymerization method. (b) Therapeutic mechanism of the Janus gel mesh for bacterial infected hernia repair. The side facing viscera has a gel structure with an anti-adhesion property, while the other side in contact with abdominal wall muscle is stimulated by bacterial infected wounds, which degrades the antifouling gel into a polymer brush that facilitates the adhesion of tissue cells, while releasing and activating bactericidal AMP to fight infection.

the normal tissue and pathological defect based on the different polymethacrylic acid (PMAA) status. Specifically, the side of JPPM in contact with abdominal wall muscle around the hernia defect would encounter multiple stimulations including strong inflammatory response and bacterial infection, which promotes the antifouling gel degradation and transforming to polymer brushes containing protonated carboxyl groups capable of facilitating the integration with tissue cells. Meanwhile, the bactericidal AMP could be rapidly released and activated to combat the infection. Of note, the side of JPPM facing viscera could maintain the non-uniform status of PMAA so that the proteins responsible for cellular adhesion and inflammatory response was effectively resisted, demonstrating a robust inhibition for the abdominal adhesion.

2. Experimental section

2.1. Materials

Benzophenone (BP), methacrylic acid (MA) and bis[2-(methacryloyloxy)ethyl] phosphate (BMP) and alkaline phosphatase (ALP) were obtained from Aladdin (Shanghai, China). 2-Hydroxy-1-(4-(2-hydroxyethoxy)phenyl)-2-methylpropan-1-one (I2959) and methylene blue was purchased from Innochem (Beijing, China). Antimicrobial peptide (AMP, HHC36:KRWKWWRR) was received from LookChem Co. Ltd, China. Fluorescein-conjugated bovine serum albumin (FITC-BSA) were purchased from BIOS (Beijing, China). Gram-positive *Staphylococcus aureus* (G^+ *S. aureus*; ATCC 6538), Gram-negative *Escherichia coli* (G^- *E. coli*; ATCC 25922), luria-bertani (LB) broth, and phosphate buffered solution (PBS; 0.1 mol/L, pH 7.4) were obtained from Dingguo Biotechnology (China). Cell Counting Kit-8 (CCK-8) was purchased from Boster Biological Technology Co. Ltd. Sprague-Dawley (SD) rats (210–250 g) purchased from basic medical college of Jilin University were used for in vivo assay. PP mesh (mono-filament, 50 g/m², pore size of 1.8 mm) was kindly provided by WEGO Co. Ltd, China. Unless otherwise stated, all chemical reagents and solvents were used directly as received.

2.2. Immobilization of BP on PP film

Flat PP film was initially chosen as a model substrate for various performance evaluations in vitro. PP (1.0 × 1.2 cm²) was ultrasonically cleaned and thoroughly dried. Then, they were treated with oxygen plasma for 3 min, immersed in ethanol containing BP (10 wt%) for 30 min. These samples (denoted as PP-BP) were washed with ethanol thoroughly and dried under N₂.

2.3. Preparation of gel surfaces

To conduct photo-initiated graft polymerization, the solutions containing functional monomers, crosslinking agent, AMP and initiator in 40 v/v% methanol aqueous solution were degassed with nitrogen stream before use. Then, 30 μL of above solution was added onto PP-BP surface and covered with quartz plate, followed by exposing to UV light (λ = 380 nm) for 6 min. The obtained sample was rinsed with water and dried. Different contents of crosslinking agent BMP (0.5 wt%, 1 wt%, 5 wt%) are used to explore the optimal gel coating ratio. It is worth noting that the above operations are repeated on both sides of the PP film.

The MA of modified gel surfaces was visualized by methylene blue staining. The samples were immersed into 1 mL of 20 mg/mL MB solutions for 30 min, and then were rinsed with water and dried.

2.4. Surface characterization

The chemical structures of as-prepared samples were analyzed using attenuated total reflection Fourier transform infrared spectroscopy (ATR-FTIR, BRUKER Vertex 70). The surface composition was detected by X-ray photoelectron spectroscopy at room temperature (XPS, VG

Scientific ESCA MK II Thermo Advantage V3.20 analyzer equipped with an Al K α anode mono-X-ray source ($h\nu$ = 1486.6 eV). The atomic concentrations of the elements were calculated by the peak area ratios. Water contact angles (WCAs) were measured by a sessile-drop method using a 2 μL droplet of deionized water (KRÜSS GMBH, Germany). The surface morphology of gel mesh was tested using scanning electron microscope (SEM, America).

2.5. Verification of phosphatase responsiveness of gel surface

In order to verify the bacterial phosphatase responsiveness of gel coating, the samples were incubated with LB suspension containing 10⁶ CFU/mL of bacteria (*S. aureus* or *E. coli*) for a certain period of time. The samples were taken out and repeatedly cleaned overnight with 5 wt% SDS solution and ultra-pure water, followed by drying before use. Samples not cocultured with bacteria were used as controls. XPS was used to detect the P 2P spectrum and element content of the samples.

2.6. Evaluation of HHC36 AMP responsive release

High performance liquid chromatography (HPLC) was used to evaluate HHC36 release. Two piece of modified samples (1.0 × 1.2 cm²) were immersed into 1 mL of PBS (pH = 5) and PBS (pH = 7.2) at 37 °C for a desired time. At the specified time interval, 1 mL of each solution was taken at 280 nm for peak area analysis. The release is measured using the HHC36 calibration curve. For reference, MAP samples were pretreated with ALP aqueous solution for 2 h.

2.7. Evaluation of antifouling properties

The representative fluorescent protein FITC-BSA was used as the model protein to detect the antifouling performance (protein adsorption) as described previously [29]. First, PBS solution containing 50 μg/mL of FITC-BSA as working solution was prepared. 500 μL of above-mentioned solution was added on the sample surface and incubated at 37 °C for 2 h in dark. Afterwards, the sample was then gently rinsed with PBS and water to remove excess fluorescent proteins and freeze-dried. Laser confocal microscope (CLSM) was then used to detect the fluorescence intensity of the surface at 488 nm excitation wavelength, and ZEN 2011 software was used to photograph and analyze the images.

Cell adhesion was tested to further evaluate the effect of Janus gel mesh on cell behaviors by using live/dead assay. All samples were put into 24-well plate and L929 fibroblasts at a density of 5 × 10⁴ cells per well were seeded on the gel surface. After 24 h incubation, the samples were carefully washed by PBS for three times, Calcein-AM solution was added to stain the cells on the samples for 20 min at 37 °C and the stained cells were observed under a CLSM.

2.8. Responsive antibacterial activities in vitro

S. aureus and *E. coli* were used as representative bacterial strain to evaluate the responsive bactericidal performance of gel mesh surfaces according to our previous reports [30–32]. Prior to test the pH and phosphatase responsive characteristics of the gel surface, the bactericidal contact time of 4 h and 8 h was selected to evaluate the ability of releasing antimicrobial peptides to kill bacteria. First, the bacterial strains on agar plates were transferred to LB liquid medium and shaken for 12 h on a 37 °C constant temperature shaker. The bacterial solution was centrifuged at 3000 rpm for 7 min. The supernatant was poured out, and the bacterial remains were removed by repeated cleaning with PBS for 3 times. Finally, 10⁸ CFU/mL of bacterial solution was prepared by using PBS, followed by diluting the abovementioned solution into 10⁵ CFU/mL by using LB liquid medium.

The samples (1.2 × 1.5 cm²) sterilized by ultraviolet light for 30 min were placed in a 12-well plate, followed by adding 10 μL of bacterial suspension (10⁵ CFU/mL) on their surfaces for co-culture in a 37 °C

incubator for 4 h and 8 h, respectively. First, polypropylene (PE) film was covered ($1.0 \times 1.0 \text{ cm}^2$) onto each sample surface to ensure the evenly spread of bacterial suspension. The samples were then immersed into 2 mL PBS solution and treated with ultrasound for 3 min to remove the adherent bacteria. Finally, 100 μL of each diluted bacterial suspension was plated on the pre-prepared LB agar plates and incubated at 37°C for 18 h. The bactericidal activity was assessed by calculating the number of viable bacterial cell colonies. The log reduction in the number of visible bacteria (reduction of log scale) was calculated to evaluate the bactericidal efficiency using the equation:

$$\text{Log bacterial reduction} = \log A - \log B \quad (1)$$

where A is the bacterial number on PP samples and B is the bacterial number on gel coating samples.

In addition, the morphology of bacteria adhered to the sample surface was further observed. After contact with bacteria, the samples were rinsed with ultrapure water and fixed with 4 % paraformaldehyde for 4 h. After freeze-drying, the morphology of adhered bacteria was analyzed by SEM. The LIVE/DEAD BacLight Viability Kit was employed to evaluate the viability of bacteria. In the experiment, gel surfaces exposed to bacteria (4 h and 8 h) were gently rinsed, and 30 μL solution containing SYTO9 (3.34 mM) and PI (20 mM) (1:1, v/v) was dropped onto the surface and stained the bacteria for 15 min in dark. After washing and drying, the live/dead state of the bacteria was observed by CLSM.

2.9. Hemocompatibility and cytocompatibility assay

The hemolytic potentials of the modified PP samples were examined according to our previous work [29,33]. Fresh blood was extracted from a healthy rabbit in accordance with the guidelines issued by the Animal Ethics Committee of Changchun Institute of Applied Chemistry, Chinese Academy of Sciences. Rabbit red blood cells (RBCs) were separated from the whole blood by centrifuging at 1000 rpm for 15 min. 300 μL of freshly diluted RBCs suspended in saline solution (5 v/v%) was dropped onto each sample surfaces and co-incubated for 2 h at 37°C . RBCs dispersed in ultrapure water and normal saline solution were taken as the positive and negative controls, respectively. Then, the solutions on the surfaces were transferred to the centrifuge tubes and centrifuged at 3000 rpm for 3 min. The absorbance of the released hemoglobin in each supernatant was measured by using a microplate reader (TECAN SUNRISE, Swiss) at 540 nm. Hemolysis percentage was calculated according to the following formula:

$$\text{Hemolysis}(\%) = \frac{OD_{\text{sample}} - OD_{\text{negative control}}}{OD_{\text{positive control}} - OD_{\text{negative control}}} \times 100\% \quad (2)$$

where OD_{sample} , $OD_{\text{positive control}}$ and $OD_{\text{negative control}}$ represented the absorbance values of the modified sample, the positive control (ultrapure water) and the negative control (normal saline), respectively.

The CCK-8 assay was conducted to test the cytotoxicity of the gel coating samples according to ISO 10993-5 standards. L929 murine fibroblasts cells at a density of 3×10^4 cells were seeded in a 48-well plate, and incubated in a humidified atmosphere of 5 % CO_2 at 37°C for 24 h. Then, the different PP samples were placed on the top of the cell layer. The parallel experiment without sample was carried out as a control. The activity of the cells was studied using the conventional CCK-8 method. After 18 h of incubation at 37°C , the samples and culture medium were removed. CCK-8 solution was mixed with culture media at a volume ratio of 1:10 to obtain CCK-8 containing media. The culture media in the wells were replaced with 200 μL of culture media containing CCK-8, and incubated at 37°C for 1 h. The $OD_{450 \text{ nm}}$ value was measured by using the microplate reader. The results were expressed as percentages relative to the OD value obtained in the control group. The cytocompatibility of gel mesh was tested by using live/dead assay. After the gel mesh was put in the wells, L929 fibroblasts were cultured on samples for 24 h. Calcein acetoxymethyl ester (calcein-AM) solution was

added to stain the cells on the samples for 20 min at 37°C and the stained cells were observed by CLSM.

2.10. Construction of bacterial infected hernia repair on the rat model

This experiment is based on the relevant requirements of the Chinese Academy of Sciences Experimental Animal Care Guidelines. All animal experiments involved in this paper have been approved by the Animal Ethics Committee of Changchun Institute of Applied Chemistry, Chinese Academy of Sciences. *S. aureus* was taken as the representative strain for infection experiment. In detail, 25 μL of *S. aureus* suspended in LB (10^6 CFU/mL) was dropped on each sample and dried on a clean bench to prepare the bacteria-contaminated samples. The gel samples were used as experimental group, and PP samples were used as control group. According to experimental requirements, SD rats were chosen as the animal model of abdominal wall hernia infection. Six adult female SD rats (~200 g) were anesthetized with isoflurane, followed by shaving their abdominal hair and iodophor disinfecting the skin. Then the abdomen was divided into two parts along the midline, and a full-thickness abdominal wall defect with a diameter of 15 mm was made by removing the abdominal wall tissue. The bacterium-contaminated samples were implanted into the defect site (the infected side was in contact with the abdominal cavity) and closed with 4-0 polyglycolic acid suture.

The rats were sacrificed on days 3. The abdominal wall was opened for observation, and the tissues around the samples were harvested for evaluation of bacterial colonies and histological observation via H&E staining.

2.11. Evaluation of tissue adhesion on the rat model

All operations on the animals were carried out in accordance with the policies of Changchun Institute of Applied Chemistry, Chinese Academy of Sciences. A SD rat model of abdominal wall defect was performed to evaluate the degree of tissue adhesion prevented by modified mesh. The rats were anesthetized with isoflurane in oxygen and maintained on isoflurane in oxygen via a nose cone. After the abdominal skin was sterilized with iodophor, the abdomen was draped with a sterile paper drape and a 3 cm ventral mid-line incision was made in the central third of the abdomen. The abdomen skin was incised, and a defection with the diameter of 15 mm was made on the muscle. Then the meshes were fixed to the corresponding abdominal wall with 4-0 polyglycolic acid sutures. At 14 days, the rats were sacrificed. The abdomen is opened and the entire abdominal cavity is thoroughly explored macroscopically to identify the degree of adhesion. The pathological changes of the abdominal wall after surgery were observed by histopathological examinations, including hematoxylin and eosin (H&E) and Masson's trichrome staining.

2.12. Statistical analysis

In all experiments, at least 3 different parallel samples were selected. And each assay was carried out for at least 3 times to ensure the dependability of the result. The report data were the mean values of triplicate specimens. Statistical analysis was performed by the analysis of variance (ANOVA) method. In all evaluations, $p < 0.05$ was considered as statistically significant, and the data were indicated with (*) for $p < 0.05$, (**) for $p < 0.01$, and (***) for $p < 0.001$, respectively.

3. Results and discussion

3.1. Determination of composition of gel surfaces

The bacterial pH/ phosphatase dual-responsive gel coating onto PP was constructed by photo-initiated graft polymerization method. Initially, the initiator BP was pre-adsorbed on the surface of PP for UV

light-initiated graft polymerization of MA and BMP, during which antimicrobial peptide HHC36 could be one-step encapsulated in the hydrogel. The BMP as a crosslinking agent also has the characteristics of phosphatase responsiveness. To confirm the optimal BMP content for phosphatase responsiveness, the different BMP concentration was used for polymerization to prepare the gel surface, denoted as MAP0.5 (containing 0.5 wt% of crosslinker BMP), MAP1 (containing 1 wt% of crosslinker BMP) and MAP5 (containing 5 wt% of crosslinker BMP) and the changes of P 2p spectrum before and after the surfaces contacting with bacteria were detected by XPS (Figure S1). In the P 2p core-level spectra, all gel samples showed a strong P 2p characteristic peak before contacting with bacteria. But after co-incubation with *S. aureus* and *E. coli*, the P 2p peak on MAP0.5 and MAP1 disappeared, whereas the P 2p characteristic peak on MAP5 showed no significant change. According to the comparison of changes in surface atom composition (Table S1), the content of P element on MAP0.5 and MAP1 was 0.09 %

and 0.16 % respectively, and decreased to 0 % after co-incubation with bacteria, indicating that bacterial phosphatase could effectively break the phosphate ester bond. However, the same for MAP5 surface was basically unchanged at the same conditions, which may be due to the fact that when the same amount of phosphatase secreted by bacteria was the same, the cross-linking density of gel coating increased with the increase of BMP amount, leading to a decrease in the degree of damage to the gel coating.

To further verify our hypothesis, *E. coli* was selected as a representative strain to conduct a preliminary evaluation of the bactericidal performance of the surface. 10 μL of 10^5 CFU/mL LB bacterial droplets were added to the sample surface and incubated in a 37 °C incubator for 8 h for colony count. As shown in Figure S2, all the modified samples showed responsive bactericidal properties, but MAP1 sample showed the best bactericidal effect, which can be explained by the balance between AMP release and bactericidal effect when the gel-coating

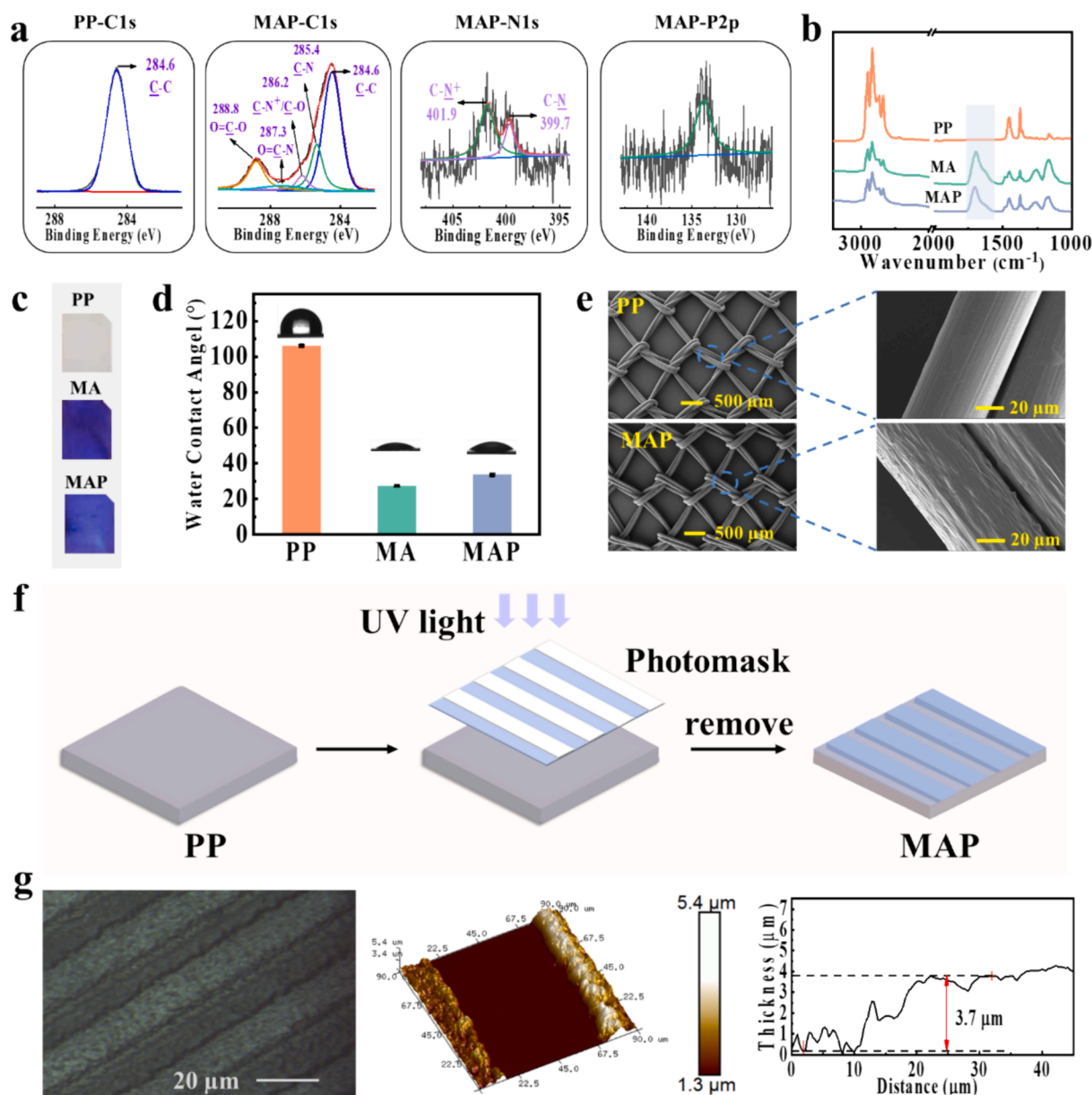


Fig. 1. Surface characterization of the gel coating. (a) Core-level spectra of C 1 s, N 1 s, and P 2p from different flat samples. (b) ATR-FTIR spectra of various flat sample surfaces. (c) Images of methylene blue staining. (d) Water contact angles of flat PP, MA and MAP. (e) SEM images of the surface of various meshes of gel coating. (f) Schematic procedure for construction and indication of patterned MAP surface. (g) Optical images and AFM images of patterned flat MAP surface. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

crosslinker content was 1 wt%. Therefore, MAP1 was selected as a representative sample (defined as MAP) for subsequent performance evaluation.

3.2. Preparation and characterization of the gel mesh

After determining the content of crosslinker BMP, the comparison sample with a 500 $\mu\text{g/mL}$ concentration of antimicrobial peptides were set up. More specifically, MA was used as the sample without crosslinker, and MAA was set as the sample with crosslinker BMA but without phosphatase response (the content of crosslinker was the same as the mole ratio of BMP). In order to facilitate the observation and characterization of gel coating composition, a flat film PP with the same composition as the mesh was used for experimental study. The successful preparation of MAP gel surface was confirmed by comparing the surface composition of bare flat PP and MAP using XPS. As can be seen from Fig. 1a, compared with the unmodified PP sample, characteristic peaks O=C-N of MAP surface stemming from antimicrobial peptides appeared at 287.3 eV in C1s core-level spectrum. The N 1s core-level spectra of MAP can be divided into two peaks, an amine group (C-N at 399.7 eV) and an ammonium ion (C-N⁺ at 401.9 eV), in which the latter was associated with the partial protonation of amino groups of antimicrobial peptides [34]. In the P 2p core-level spectra, a major peak at ~ 133.8 eV corresponding to crosslinking agent BMP was presented on the MAP surface. The FTIR spectra of the coatings is shown in Fig. 1b. The assignments corresponding to MA included the C-O antisymmetric stretch and the C-O symmetric stretch at 1258 and 1160 cm^{-1} , respectively. The C=O stretching vibration band of MA was observed at about 1704 cm^{-1} . Meanwhile the characteristic absorption peak of O-C=O corresponding to the carbonyl groups of the MAP surface at 1704 cm^{-1}

also confirmed that the gel coating is successfully constructed.

Moreover, methylene blue (MB) staining was applied to visualize the carboxyl groups on the surface. Methylene blue is a saline-based dye that strongly interacts with carboxyl groups by ion exchange [35]. MB experimental results (Fig. 1c) showed that MA and MAP absorbed a large amount of MB, presenting a dark blue color, which was due to the rich carboxyl group of MA in the gel coating, further indicating the successful grafting of MA segment. The wettability of the surfaces was studied by WCAs analysis (Fig. 1d). The surface of flat PP was hydrophobic with the WCA of $\sim 106.5^\circ$. Being modified with MA, MA surface possessed high hydrophilicity with the WCA of $\sim 27.5^\circ$. With further functionalization by BMP, the WCA values of MAP were $\sim 33.6^\circ$. The good hydrophilic properties were beneficial to improve the biocompatibility of the functionalized surfaces. Finally, the morphology of gel coating on the hernia meshes was verified by SEM. As shown in Fig. 1e, compared with the original PP mesh, the surface of the MAP sample is rougher, indicating that the gel is coated on the surface of the mesh nanofibers. To obtain the thickness of the gel coating, SEM measurements were conducted, and the results are shown in Figure S3. The gel coating with the height of 5.7 μm was obtained. Furthermore, we constructed a stripe patterned structure by a photomask method on the PP substrate which could give vivid visualization of the stripe coating thickness through evaluating the height of the stripe structure (Fig. 1f). As presented in the atomic force microscope (AFM) images, the height of the gel coating was ~ 3.5 μm (Fig. 1g). Collectively, the thickness of the gel coating can reach several micrometers.

3.3. HHC36 release profiles in vitro

The pre-prepared gel coating surface can respond to the stimulation

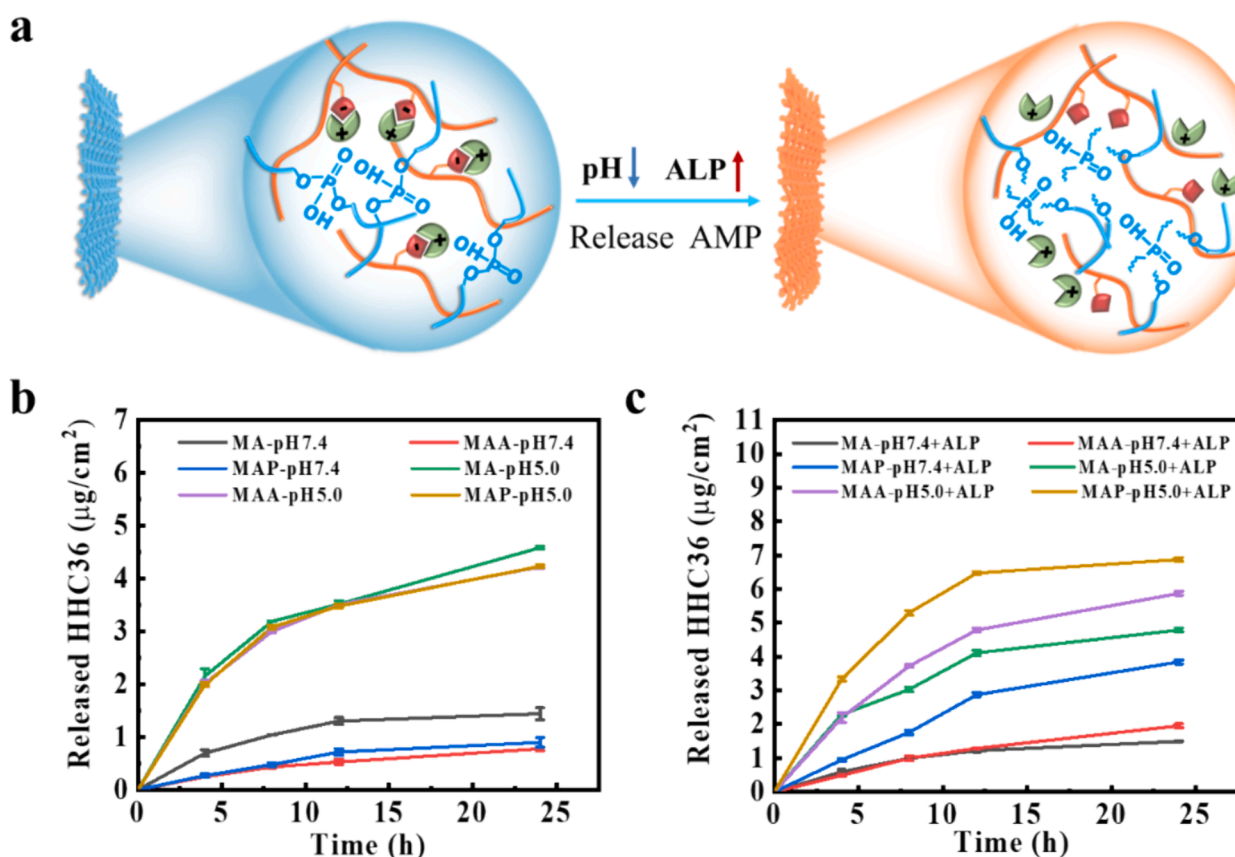


Fig. 2. (a) The side of gel coating in contact with abdominal wall muscle around the hernia defect encounter bacterial stimulations including bacterial lower pH value and bacterial alkaline phosphatase. The antifouling gel degraded and the bactericidal AMP was released. (b) and (c) Time-dependent release profile of HHC36 under different conditions.

of both bacterial pH and phosphatase (ALP), resulting in the more sensitive release of HHC36 via simultaneous weaken the electrostatic interaction between MA and HHC36 and break the phosphodiester linkage (Fig. 2a) [36–39]. In order to prove the characteristics of MAP autonomously responding to bacterial phosphatase, MAP was incubated with *S.aureus*, *E.coli* and alkaline phosphatase solutions respectively, and the changes of P 2p spectrum of the surface before and after contact with the solution were detected by XPS. As shown in Table 1, in the P 2p nuclear level spectrum of XPS, MAP samples showed the disappearance of P elements after contact with *S.aureus* and *E.coli*, and the reduction of P elements from 0.16 to 0.08 after contact with ALP. These results further demonstrate that MAP facing abdominal wall muscle around the hernia defect can spontaneously respond to changes in the bacterial microenvironment.

Fig. 2b and Fig. 2c illustrated the release curve of HHC36 under different stimuli. In particular, MA samples were selected to compare the effect of crosslinking agents on the release of HHC36, and MAA samples (not contain phosphatase responsive crosslinking agent BMP) were chosen to compare the effect of dual responsive characteristics on the release of antimicrobial peptides. At pH = 7.4, the release amounts of HHC36 on MAP and MAA were about 0.4 $\mu\text{g}/\text{cm}^2$ within the initial 4 h, while MA released 1.0 $\mu\text{g}/\text{cm}^2$ of HHC36 (Fig. 2b), indicating that the crossing-linking agent in gel could slow down the release of antimicrobial peptides under physiological conditions. In contrast, in a simulated condition of bacterial infection, MA and MAA released $\sim 2.2 \mu\text{g}/\text{cm}^2$ within 4 h in the pH 5.0 and ALP treatments, respectively (Fig. 2c). Obviously, MAP was able to reach the release value of $\sim 3.3 \mu\text{g}/\text{cm}^2$, showing fast response to the pH stimulus. In the case of MAP at pH 5.0 without ALP pretreatment, a maximal release dosage of HHC36 was detected, but it was only $\sim 4.2 \mu\text{g}/\text{cm}^2$. These results suggested that MAP had a phosphatase and pH-dependent antimicrobial peptide release profile and therefore could be activated with appropriate stimulation to accurately combat complex bacterial infection problems. It should be noted that release of antimicrobial peptide HHC36 from the gel surface is highly conducive as release of large quantities of antimicrobial agents in a short period of time could rapidly eradicate bacteria existing around the surface of medical devices.

3.4. Anti-adhesive behaviors of the Janus gel coating

To evaluate the effect of Janus gel mesh on anti-adhesive behaviors, biological adhesion assessments were performed in vitro (Fig. 3a). Adsorption of proteins on the surface of biomaterials is considered to be the first step in a cascade of surface adverse reactions such as thrombosis, immune responses, and infection [40,41]. Medical devices are usually in a protein-rich environment. Therefore, it is necessary to evaluate the anti-protein adhesion properties of biomaterials. Herein, FITC-BSA was selected as the representative protein for evaluating on the protein resistance of the gel surfaces. As showed in Fig. 3b, the flat PP surface showed a large amount of green fluorescence, indicating that a large number of proteins were adsorbed on its surface. However, there were almost no green fluorescence on the pristine MAP surface, displaying that gel coating endowed the surface with excellent anti-protein adhesion property. As shown in Fig. 3d, the MAP surfaces showed about 8 times decrease in the fluorescence intensity, as compared with that of

Table 1

The comparison of surface atom composition changes through XPS after different surface treatments (MAP samples were exposed to *S.aureus*, *E.coli* and ALP).

Name code	Composition (at. %)			
	C	O	N	P
MAP	73.06	26.08	0.7	0.16
MAP- <i>S.aureus</i>	80.06	19.59	0.35	–
MAP- <i>E.coli</i>	79.27	20.49	0.24	–
MAP-ALP	80.18	19.22	0.52	0.08

flat PP. As fibroblasts were the main component of connective tissue during wound healing, L929 fibroblasts were selected to seed on different surfaces of gel meshes. Much more L929 fibroblasts could adhere and grow on the PP sample (Fig. 3c), while only few L929 fibroblasts were found on the surface of Janus gel sample MAP in contact with viscera. These results demonstrated the efficient regulation of Janus gel surface on biological adhesion properties of MAP. The anti-adhesion behavior of the gel coating can be interpreted that the affiliated carboxyls of the PMAA would stain completely negatively charged due to the coupling effect of positively charged HHC36, and the hydration layer is formed on the surface of the mesh to hinder intermolecular interactions between abdominal viscera and meshes, leading to the adhesion failure.

3.5. Bactericidal assay of the Janus gel coating

Under normal physiological conditions, the HHC36 could be stable loaded in the gel surface, whose release could be activated in presence of bacteria. Bacterial reproduction will lead to the decrease of acidity of local microenvironment and the increase of phosphatase concentration [42,43], which is able to trigger the rupture of phosphate ester bond of crosslinking agent BMP, release antimicrobial peptides and effectively combat bacterial infection. The loaded HHC36 could be on-demand released into the medium to kill bacteria and inhibit bacteria growth. The bactericidal property of the modified PP flat film was evaluated via the colony forming unit (CFU), morphology (SEM) and bacterial viability (live/dead fluorescence staining) using typical *S. aureus* and *E. coli* as representative bacterial strains (Fig. 4a).

The bactericidal properties of gel coating firstly evaluated by CFU. In our experiments, bacteria were first seeded onto the samples and incubated for 4 h, allowing for bacterial settlement on the surface, followed by treating with sonication vigorously in a sterile PBS solution. The bacteria were then spread on agar plate using a pipette to calculate the log reduction number of bacteria. As shown in Fig. 4c, compared with non-biocidal PP, the MAP effectively killed *S. aureus* during 4 h incubation, eventually reaching more than ~ 2.6 log reduction in bacterial number. While the bacterial number for MA and MAA was reduced up to ~ 1.9 log and ~ 1.1 log, respectively. As for *E. coli*, MAP also exhibited the most potent bactericidal ability, showing ~ 2.1 log reduction in bacterial number with relative to ~ 0.8 log and ~ 0.5 log reductions for the MA and MAA, respectively (Fig. 4e). Considering the application of the MAP in medical devices, the coating must maintain continuous bactericidal activity. The contact time between bacteria and the surface was extended to 8 h to further evaluate the bactericidal performance of the surface. As shown in Fig. S4b and S4d, MAP can effectively kill *S. aureus* after incubating with bacteria for 8 h, reducing the number of bacteria by ~ 3 log compared with that of no bactericidal activity of PP. The number of bacteria in MA and MAA decreased by ~ 0.9 log and ~ 2.6 log, respectively. For *E. coli*, MAP also showed the best bactericidal effect, reducing bacterial number by 3.3 log as compared with MA and MAA, respectively.

To investigate why MAP exhibits the best antibacterial performance, we explored the surface morphological changes of bacteria, which was characterized using SEM. The bacteria that attached to the non-antibacterial bare PP surfaces, exhibited clear edges of the cell membrane and intact bodies (Fig. 4b, 4d, S4a and S4c). In remarkable contrast, bacteria on the MA, MAA and MAP surfaces exhibited severely damaged and collapsed cell membranes. This is because loaded antimicrobial peptides are released and could effectively bind to the bacteria, leading to the breakage of the bacterial cell membrane and the death of pathogens [44].

To further confirm the bactericidal properties, the viability of adherent bacteria was evaluated using the LIVE/DEAD BacLight staining method and observed by CLSM. After co-culturing with *E. coli* or *S. aureus* suspensions for 4 h, a great number of viable bacteria (the green fluorescence) aggregated into clusters on the pristine PP surface,

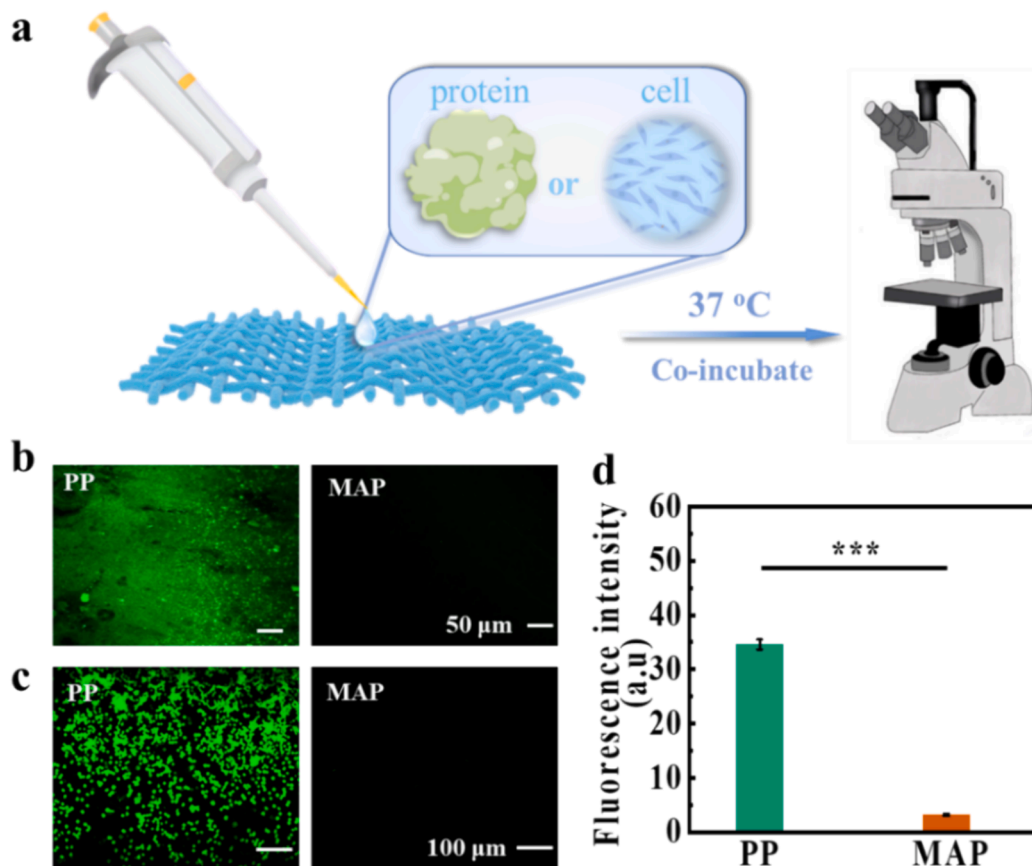


Fig. 3. (a) Schematic illustration of the adhesion behavior of proteins and cells on the flat sample surfaces. (b) Representative fluorescence microscopy images of different surfaces after adsorption of BSA proteins. (c) Fluorescence images of L929 fibroblasts cultured on the samples. (d) Fluorescence intensity statistics of various samples for adsorption of BSA proteins.

while almost no dead bacteria with the red fluorescence were observed. In contrast, MAP demonstrated intensive red fluorescence, indicating its effective killing of both types of bacteria. Although MA and MAA also exhibited good bactericidal activity, their bactericidal efficiencies were lower than that of MAP, which was consistent with bacterial counting result. MAP still showed the best bactericidal effect analyzed by the above three methods after 8 h contact with bacteria (Fig. S4a, S4c).

The above results indicated that the HHC36-loaded gel coating showed strong bactericidal capacity by combining dual response into one system. The coating is sensitive to acidity and bacterial enzymes present as triggers for HHC36 release. On the one hand, the acidic substances secreted by bacteria lead to environmental acidification, the protonation of MA in the gel weakened the electrostatic interaction of HHC36. On the other hand, bacteria secrete phosphatase as a virulence factor during invasion, which can break the phosphate ester bond in BMP and cause the decomposition of gel crosslinking network. The combination of the dual response allows the surface to accurately combat to bacterial infection, which is conducive to improving the bactericidal effect.

3.6. Hemocompatibility and cytocompatibility assay

Exceptional biocompatibility is a prerequisite for implantable mesh device applications. Thus, the hemocompatibility and cytocompatibility of the MAP mesh were evaluated *in vitro*. The hemocompatibility of the Janus gel mesh was tested by hemolysis experiment. The normal saline and ultrapure water were chosen as negative control (NC) and positive control (PC) groups, respectively. The suspension of all the sample groups displayed light red, the same as that in the negative control group, while the suspension of the ultrapure water group was dark red

and the bottom of the tube did not have precipitant. The hemolysis rate of PP, MA and MAP was 0.42 %, 0.71 %, and 0.31 %, respectively (Fig. 5a), all of which were less than 5 %, indicating their good hemocompatibility. Moreover, cytocompatibility of the meshes was further investigated by CCK-8 assay, live-dead staining and cytoskeleton staining (DAPI and TRITC-phalloidin). As exhibited in Fig. 5b, the cell viability for the MA and MAP was comparative with pure PP sample (around 100 %), indicating that the gel coating showed no cytotoxicity when the antimicrobial peptides were encapsulated in it. This result was also supported by the live/dead cell assay (Fig. 5c). In all groups of L929 cells were green and almost no dead cells were seen after incubation on 24 h. Notably, almost no cell adhesion was observed in the Janus gel mesh, which may be due to the fact that the gel coating can form a hydrated layer as a physical barrier to prevent cell adhesion. To visualize the cell viability more intuitively, the cytoskeleton staining of L929 cells on the meshes was used by DAPI and TRITC-phalloidin. L929 cells displayed normal cytoskeleton (green) and cell nuclei (blue) morphologies on all the meshes (Fig. 5d). Taken together, the good hemocompatibility and cytocompatibility made gel coating meshes safe candidate materials for hernia repair.

3.7. Anti-infection performances in an abdominal wall hernia rat model

When the abdominal wall is injured or colonized with bacterial cells, the corresponding microenvironment becomes acidic due to endogenous stimulation induced by the inflammatory response or bacterial infection. Taking advantages of the lower pH microenvironment, the JPPM side adjacent to defect of the abdominal wall could rapidly decompose and release the HHC36 to perform bactericidal function. Additionally, as reported [25], the carboxyls on the PMAA brushes would become

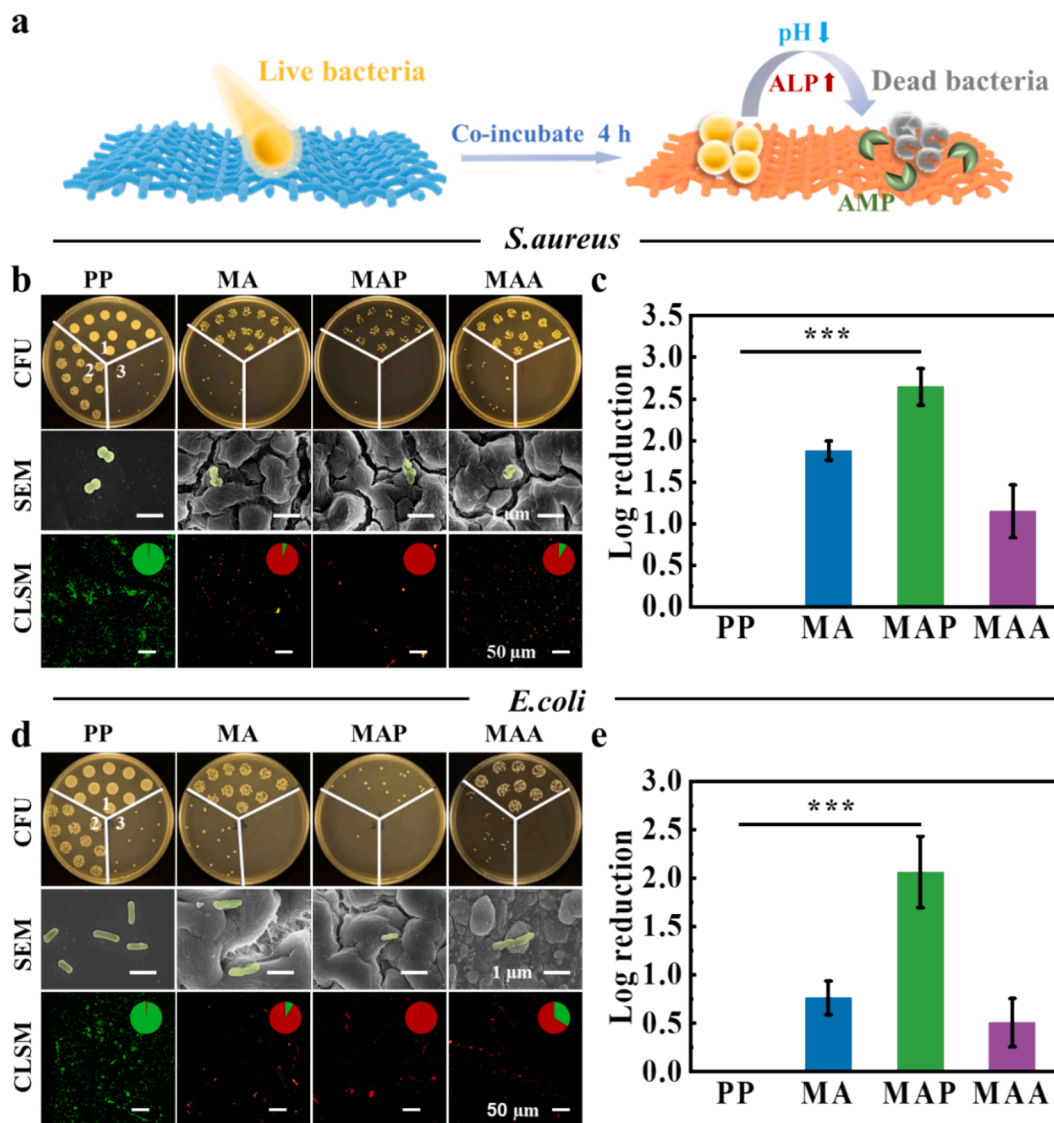


Fig. 4. Bactericidal statistics of flat samples after 4 h contact with bacteria. (a) Schematic illustration of the bactericidal activity of samples. (b) and (d) Responsive bactericidal efficacies of various surfaces evaluated by colony forming units (CFU) of residual *S. aureus* and *E. coli* on sample surfaces (1–3 represents dilution of the original bacterial solution 1, 10^2 , 10^4 respectively), morphology (The bacteria in SEM images were processed with false-color) and bacterial viability. (c) and (e) Corresponding log reductions of *S. aureus* and *E. coli* in bacterial number of various flat sample surfaces. The error bars indicated the standard deviation ($n = 3$).

nonionic and hydrophobic status locally, which may effectively promote the adhesion of cells and integration of soft tissue. In this regard, the in vivo bactericidal activity of HHC36 was firstly assessed through an animal model associated with mesh-induced infections (Fig. 6a). Due to the best in vitro bactericidal performances and good biocompatibility, MAP mesh was selected for in vivo anti-infection study, and pristine PP mesh was taken as a control. The samples were contaminated with bacterial suspension of *S. aureus* and implanted into subcutaneous abdominal wall defect of rats. At days 3, the surface of pristine PP mesh was covered with pus, indicating severe bacterial infection, while almost no residual pus existed on the MAP samples surface, which demonstrated the bacterial infection was inhibited (Fig. 6b). The samples of anti-infection experiments were observed by CFU and the number of adhered bacteria were quantified (Fig. 6d and 6e). The pristine PP samples were fouled with a great deal of bacteria, while a few bacteria existed on the MAP surface, showing high bactericidal performance with a killing efficiency of $\sim 80\%$ on the 3rd day after implantation. For histological examination, the tissues around the implanted samples were harvested and subjected to hematoxylin and eosin (H&E) staining (Fig. 6c). Significant neutrophil infiltration was observed in the PP

group, suggesting an acute inflammatory response caused by bacterial infection. MAP group showed no obvious infiltration of inflammatory cells, which proved the inhibitory effect on bacterial infection. Subsequently, the duration of mesh treatment was extended to 7 days to evaluate its anti-infection performance. As displayed in Fig. S4a, there was still a distinct abscess in the PP mesh group, whereas no signs of infection were observed in the MAP mesh group. In sharp contrast to the large number of bacteria in the PP group, almost no bacterial colonies remained in the MAP group on the 7th day after treatment (Fig. S4b and S4c). This was consistent with the H&E staining results (Fig. S4d). We speculated that such a good therapeutic effect of gel meshes within 7 days may be due to the fact that released AMP inhibited bacterial proliferation during the acute infection phase (~ 3 days) and the small amount of bacteria remaining can be effectively killed by the immune system. However, the original PP meshes showed no inhibitory effect on the initial bacteria, resulting in the inability of the immune system to effectively kill such a large number of bacteria, so infected bacteria were still colonized in the PP mesh group after 7 days. Overall, MAP meshes could kill bacteria and alleviate the infection-induced inflammatory response in vivo.

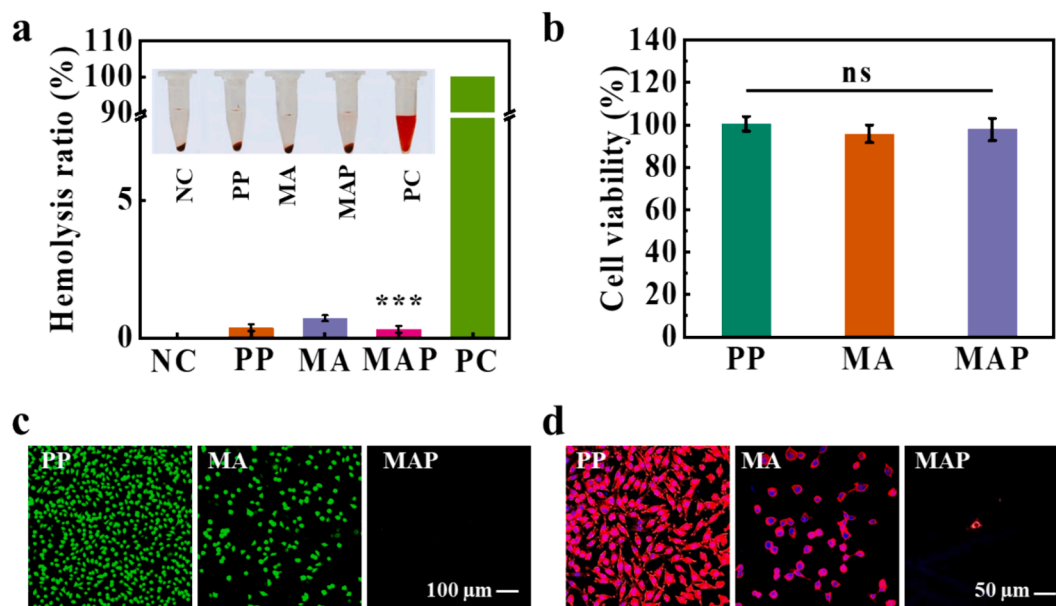


Fig. 5. Biocompatibility of various flat film samples in vitro. (a) Hemolytic activities of different samples. (b) Cell viability after being incubated with various samples. The error bars indicate the standard deviation (n = 3). (c) Live/dead staining fluorescent images of L929 fibroblasts cells after cocultivation with meshes for 24 h. Red fluorescence represents dead cells that allow propidium iodide (PI) to enter, and the green represents live cells stained with Calcein-AM. (d) Fluorescent images of L929 fibroblasts cells co-cultivated with Janus gel mesh for 24 h. Red indicates actin visualized by TRITC-phalloidin staining, and blue indicates nuclei stained by DAPI. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

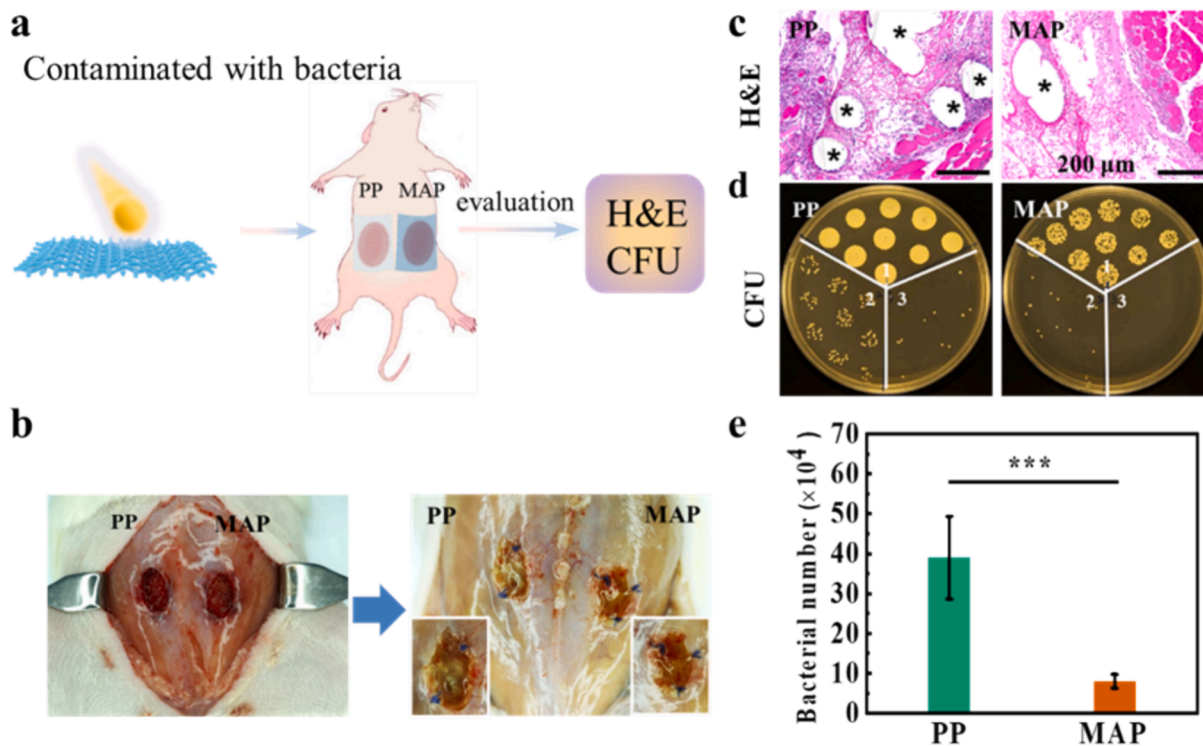


Fig. 6. (a) Schematic diagram of the infection, treatment, and monitoring regime used in SD rats with *S.aureus*. (b) In vivo evaluation of MAP as an anti-infection mesh in a rat model of abdominal wall hernia and photographs of *S. aureus*-infected hernia mesh on the rat with different treatments after 3 d therapy. (c) Representative images of H&E staining. (d) The LB-agar photographs of bacteria colonies harvested from different meshes (1–3 represents dilution of the original bacterial solution 1, 10^2 , 10^4 respectively). (e) The bacterial number of different meshes. * represents the monofilament of the polypropylene mesh.

3.8. Anti-tissue adhesion performances in an abdominal wall hernia rat model

The rat abdominal adhesion model was established in this

experiment. After 14 days of surgery, the local healing sites were examined by both gross observation and histologic evaluation (H&E staining and Masson's trichrome staining). Considering the comprehensive properties aforementioned, the MAP gel coating was finally

selected for *in vivo* anti-tissue adhesion study (Fig. 7a). The representative images of the observed adhesion formation in two groups were shown in Fig. 7b, and all groups were scored on the 14th day after implantation (Fig. 7c) according to the scoring system (Table S1) [45,46]. The PP group showed severe adhesions between the abdominal wall and cecum, with an average clinical score of 7.3 for all rats ($n = 10$). Such severe adhesions required a sharp dissection to detach the fibrous tissue surrounding both the cecum and abdominal wall. By contrast, the group with MAP meshes treatment showed almost no adhesive tissue, with an average score of 0, suggesting that the adhesions were effectively prevented. Hematoxylin and eosin (H&E) and Masson's trichrome staining were used to examine the level of inflammatory response and collagen deposition. As shown in Fig. 7d, PP-treated tissue showed a large of inflammatory cells and neutrophils in the abdominal wall tissue, while no inflammatory cell infiltration was observed in the MAP group, suggesting the process of adhesion formation is closely related to inflammation response [47–49]. In Masson staining images (Fig. 7e), abundant collagen fiber deposition formed by the proliferation of fibroblasts was present in the PP group, leading to the formation of adhesions [9]. Consistent with the H&E staining, no collagen deposition was observed in the MAP group, demonstrating a good antiadhesive effect. Based on these results, it can be confirmed that the MAP not only act as a physical barrier, but also reduce the inflammatory response in the defect tissue to prevent the occurrence of adhesion.

4. Conclusions

In summary, we have successfully demonstrated the concept that a homogeneous gel coated implant mesh can rapidly transform to a Janus mesh locally once encountering the asymmetrical microenvironments adjacent to the normal tissue and pathological defect to perform side specific functions. This endogenous stimulation-driven Janus gel coating composed of methylacrylic acid (MA), antimicrobial peptide HHC36 and crosslinking agent bis[2-(methacryloyloxy)ethyl] phosphate (BMP) was feasibly constructed by one-step photo-initiated graft polymerization method. Initially, the intact homogeneous gel structure composing of negatively charged PMAA and the positively charged HHC36 can completely inhibit the adhesion of proteins and cells on the surface, thus avoiding the adhesion of abdominal tissue. While, once encountering with the inflammatory response and bacterial infection within the defect wound site, the lower pH will weaken the electrostatic interaction between PMMA and HHC36, meanwhile, the released ALP from the bacterial metabolism could break the phosphodiester linkage from BMP, which collectively leads to the decomposition of the adjacent gel coating and the release of AMP to effectively inactivate the invaded microbes. Interestingly, due to the lower pH and the release of HHC36, the carboxyls on the PMAA brushes would become nonionic and hydrophobic status locally, which effectively promoted the adhesion of cells and integration of soft tissue. Additionally, *in vitro* and *in vivo* experiments confirmed that Janus gel mesh demonstrated outstanding biocompatibility. We believe that this novel endogenous stimulation-driven Janus

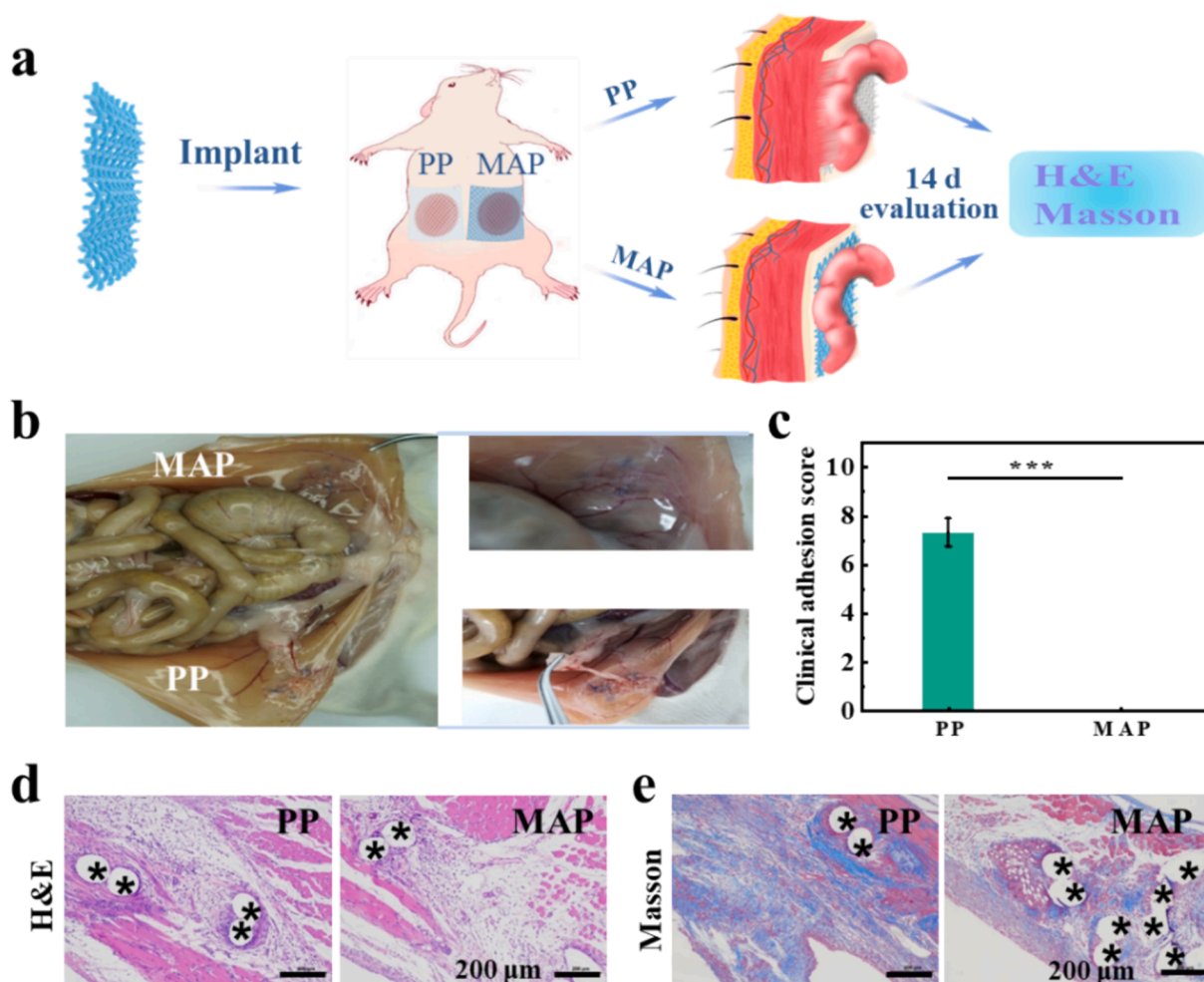


Fig. 7. (a) Schematic diagram of evaluation of adhesion resistance of different meshes after 14 d therapy. (b) Photographs of hernia mesh on the rat with different treatments. (c) Quantitative scoring analysis of adhesion. (d) Representative images of H&E staining of the tissues around the samples. (e) Images of Masson staining of the tissues around the samples. * represents the monofilament of the polypropylene mesh.

gel network provides a new approach for the design and study of novel soft tissue repair biomaterials.

CRediT authorship contribution statement

Liwei Sun: Writing – original draft, Project administration, Methodology. **Xu Zhang:** Validation. **Jiteng Zhang:** Supervision. **Xinmeng Li:** Supervision. **Jie Zhao:** Writing – review & editing. **Rujian Jiang:** Writing – review & editing, Supervision. **Lingjie Song:** Writing – review & editing, Conceptualization. **Shifang Luan:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2024.152764>.

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